AgroBot

Smart Soil Moisture Monitoring System

Project Report

IoT-Based Real-Time Soil Moisture Detection & Monitoring

NodeMCU V3 (ESP8266) | Real-Time Wi-Fi Data | Live Web Dashboard | Smart Buzzer Alerts

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Chapter 1 — Introduction

Water is vital for plant growth, yet irrigation is often poorly managed across Indian agriculture. Both over-irrigation and under-irrigation lead to significant crop losses in regions like Maharashtra and Madhya Pradesh. Farmers routinely overwater during monsoons — causing root diseases — or under-irrigate in summers, leading to crop failure. A core issue is the lack of affordable, real-time soil moisture monitoring, forcing farmers to rely on guesswork rather than data.

AgroBot addresses this directly using the Internet of Things. A soil moisture sensor is connected to the internet via a NodeMCU V3 microcontroller, which reads moisture levels every 10 seconds and transmits the data wirelessly to a web server. Users can then monitor soil conditions in real time from any phone or laptop — without visiting the field.

The system is built around a single NodeMCU V3 board (ESP-12E / ESP8266), which handles both sensor reading and Wi-Fi data transmission. A piezoelectric buzzer provides immediate local alerts even when there is no internet connectivity. The companion web dashboard displays the live moisture reading and a 7-day trend chart.

Chapter 2 — Objectives & Scope

Objectives

- Design and assemble a single compact IoT device using NodeMCU V3, a resistive soil moisture sensor, and a piezoelectric buzzer — all enclosed together as one unit.
- Write and upload firmware to the NodeMCU to read analog moisture values via the on-chip ADC, convert them to a moisture percentage, and trigger buzzer alerts when moisture leaves the safe range.
- Transmit moisture readings wirelessly to a web server using the NodeMCU built-in Wi-Fi and HTTP POST requests.
- Build a web dashboard displaying the live soil moisture reading and a weekly trend chart showing the past seven days.
- Provide immediate local buzzer alerts (independent of internet connectivity) when soil conditions are critical.

Scope

The project covers the complete design, development, and testing of a single-node soil moisture monitoring prototype. The hardware circuit, firmware, buzzer alert logic, and physical enclosure are complete. The website front-end is done. Server-side integration is being finalised. Multi-node support, an automated drainage system, and a farm security system are planned for future work and are outside the current scope.

Chapter 3 — Hardware Revision

The original mid-semester design used two boards — an Arduino Uno microcontroller and a separate ESP8266 ESP-01 Wi-Fi module — communicating over UART using AT commands. This required carefully matched baud rates, cross-wired TX/RX pins, and a voltage divider between the Arduino 5V TX and the ESP8266 3.3V RX pin. In practice, the AT commands frequently failed, making the system unreliable and difficult to debug.

The solution was to switch to a single NodeMCU V3 board (ESP-12E). This board has Wi-Fi built in and connects directly to a laptop via Micro USB. All AT-command complexity was eliminated — Wi-Fi is managed in a single *WiFi.begin()* call using the *ESP8266WiFi.h* library.

Feature	Mid-Semester Design	Final Design
Main board	Arduino Uno (ATmega328P)	NodeMCU V3 (ESP-12E)
Wi-Fi	Separate ESP8266 ESP-01 module	Built into NodeMCU
Board-to-board comms	UART AT commands	Not needed
Voltage divider	Required (5V → 3.3V)	Not needed
USB driver	Comes with Windows	CH340C driver required
Wi-Fi code	Manual AT command strings	WiFi.h library
Number of boards	2	1
Reliability	AT commands sometimes failed	Stable

Chapter 4 — Components & Technical Description

NodeMCU V3 (ESP-12E) Microcontroller

Chip	ESP8266 (ESP-12E module)
CPU	32-bit, 80 MHz
Wi-Fi	802.11 b/g/n — built in
USB chip	CH340C — direct Micro USB connection
ADC	1 channel (A0), 10-bit (0–1023)
Programming	Arduino IDE — NodeMCU 1.0 (ESP-12E) board profile
Libraries used	ESP8266WiFi.h, ESP8266HTTPClient.h

Resistive Soil Moisture Sensor

Type	Resistive — two metal probes + control board
Principle	Wet soil conducts more current (lower resistance); dry soil less (higher resistance)
Output	Analog voltage → NodeMCU pin A0
ADC mapping	1023 = 0% moisture (fully dry) · 0 = 100% moisture (fully wet)
Noise filtering	5 readings averaged per cycle with 50 ms gaps

Piezoelectric Buzzer

Connection	Positive leg → NodeMCU pin D2
Dry alert	3 short beeps (ADC below DRY_THRESHOLD)
Wet alert	2 long beeps (ADC above WET_THRESHOLD)
Safe range	Buzzer off
Offline capable	Runs entirely on NodeMCU — no internet required

Chapter 5 — System Design & Methodology

Firmware Flow

- **Step 1:** NodeMCU powers on and initialises the ADC (pin A0) and buzzer (pin D2).
- **Step 2:** Every 10 seconds, pin A0 is sampled 5 times with 50 ms gaps; the average is computed.
- **Step 3:** The raw ADC value (0–1023) is mapped to a moisture percentage using map() and clamped with constrain().
- **Step 4:** The raw value is compared to DRY_THRESHOLD and WET_THRESHOLD constants — buzzer triggers if outside the safe range.
- **Step 5:** NodeMCU connects to Wi-Fi via WiFi.begin() and sends the moisture percentage to the server as an HTTP POST request.
- **Step 6:** The loop waits 10 seconds and repeats from step 2.

Wiring Summary

Component	NodeMCU Pin	Function
Soil Moisture Sensor (AO)	A0	Analog moisture voltage input
Buzzer (+)	D2	Digital alert output
Power	Micro USB	5V supply → onboard 3.3V regulator

Chapter 6 — Hardware Setup

The photograph below shows the assembled AgroBot prototype. The NodeMCU V3 board is visible at the top right, with the soil moisture sensor's control board connected on the left side and its probe pair inserted into the soil. The piezoelectric buzzer is the circular black component at the bottom, connected to pin D2 via the purple wire.

Figure 1 — AgroBot prototype: NodeMCU V3 (top right), soil moisture sensor (left), piezoelectric buzzer (bottom). Sensor probes are inserted directly into soil.

Chapter 7 — Web Dashboard

The AgroBot web dashboard is deployed on Vercel and accessible at **kanishkdubey1946-del-agobot-smart-i.vercel.app**. It is built with HTML, CSS, and Chart.js, and features a dark green theme consistent with the AgroBot brand. The dashboard comprises four main sections:

- Live moisture gauge — circular indicator showing the current soil moisture percentage in real time.
- Quick stats row — Today's average, lowest, and highest moisture readings at a glance.
- Local weather panel — Temperature, humidity, wind speed, and conditions fetched via location access.
- Moisture Trends chart — Daily average or hourly Chart.js graph showing moisture patterns.
- Alert History — Log of buzzer-triggering events (dry/wet threshold breaches).
- Device Information card — Displays device name, ID, microcontroller type, sensor type, location, and server URL.

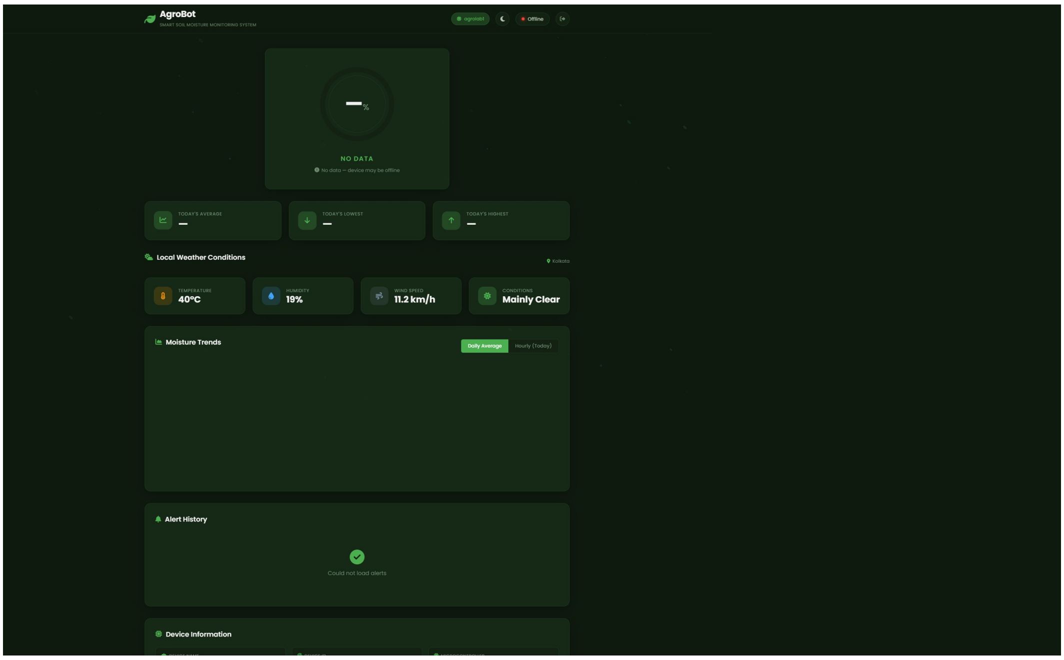

Figure 2 — AgroBot monitoring dashboard showing the moisture gauge, quick stats, local weather panel, moisture trends chart, alert history, and device information.

Chapter 8 — Device Connection (Login Page)

Before accessing the dashboard, users connect their NodeMCU device through the login page. This page accepts the device name, a unique device ID, the server IP or URL where the NodeMCU is sending data, and the Wi-Fi module type (ESP8266 NodeMCU, ESP32, ESP8266-01, or Other). An optional access key provides additional security. The page also supports geolocation to enable local weather data on the dashboard.

Previously connected devices are saved locally and listed as quick-connect tiles below the form, so users do not need to re-enter credentials each time. The connection is local — no cloud account is required.

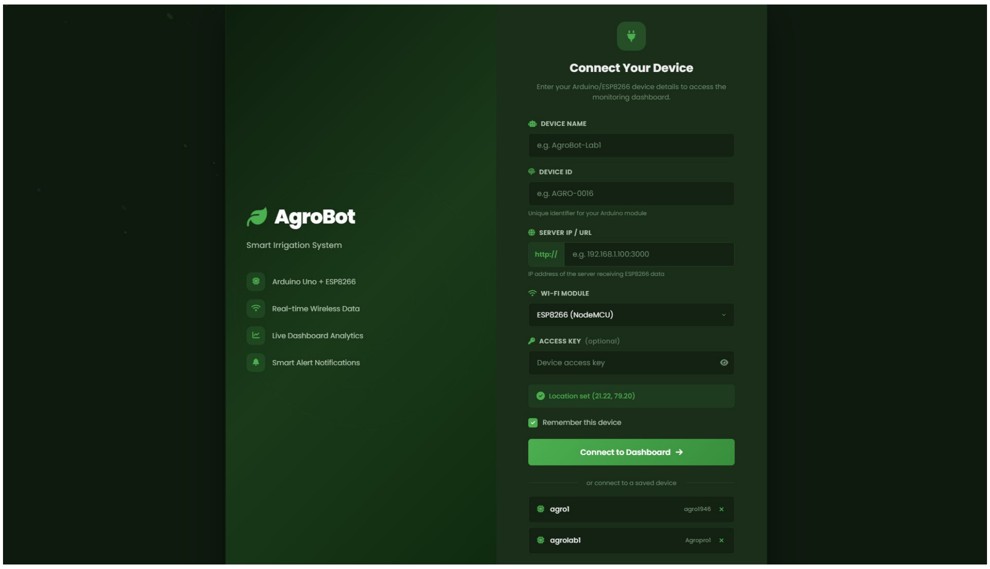

Figure 3 — AgroBot device connection page. Users enter Device Name, Device ID, Server IP/URL, and Wi-Fi module type to link their NodeMCU to the dashboard.

Chapter 9 — Current Progress

Component	Status	Notes
Hardware assembly & enclosure	■ Complete	NodeMCU V3, sensor, buzzer wired and enclosed
Firmware upload & sensor reading	■ Complete	Serial Monitor confirms readings every 10 s
Buzzer alert logic	■ Complete	Dry/wet thresholds verified by test
Website front-end	■ Complete	Dashboard + login page live on Vercel
Server-side API & database	■ In Progress	HTTP POST endpoint being finalised
Full end-to-end test	■ Pending	Awaiting server-side completion

Chapter 10 — Conclusion & Future Work

AgroBot has made concrete, measurable progress. The hardware went through a significant redesign — moving from a two-board Arduino Uno + ESP8266 setup to a single, more reliable NodeMCU V3 board. The firmware is stable: the sensor reads moisture correctly every 10 seconds and the buzzer triggers the correct alert patterns for both dry and wet conditions, verified through the Serial Monitor. The web dashboard front-end is complete and deployed.

The remaining work is completing the server-side API so the NodeMCU HTTP POST reaches the database and the dashboard shows live data, followed by a full end-to-end integration test.

Planned Future Work

- **End-to-end server integration** — Connect NodeMCU HTTP POST to the live database and wire API endpoints to the dashboard.
- **Real-farm deployment** — Test the device across different soil types under actual field conditions.
- **Automated drainage system** — A relay-controlled valve that opens automatically when moisture exceeds the wet threshold.
- **Farm security system** — HC-SR04 ultrasonic sensors along the farm perimeter to detect animal or human intrusion.
- **Multi-node support** — Monitor multiple sensor nodes from different field locations on one unified dashboard.

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